Create Frame-by-Frame Animation Using Characters

Project Presentation

Semester: 1st

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Subject: Programming Fundamental

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CHAPTER 1

INTRODUCTION:

Animation is an important concept in computer programming that helps in visualizing movements and real-time behavior of objects. This project is a **Console-Based Animation System** developed using the **C++ programming language**.
The program uses **ASCII characters** to create an animated scene inside the console window.

The animation includes multiple moving elements such as sun, clouds, birds, mountains, trees, house, windmill, train, car, walking man, river, fish, and grass. These objects move frame by frame using loops, conditional statements, and time delay.

This project helps beginners understand how animation logic works without using graphics libraries.

CHAPTER 2

OBJECTIVES:

The main objectives of this project are:

- To understand the concept of **frame-by-frame animation**
- To implement animation using **C++ console**
- To practice **loops, functions, and conditions**
- To learn how to control speed using the Sleep() function
- To improve logical and problem-solving skills
- To create an **interactive menu-driven program**

CHAPTER 3

KEY FEATURES:

- Menu-driven system (Start / Exit)
- Multiple animated objects in a single scene
- Independent movement speed for each object
- Use of ASCII art for animation
- Blinking house lights effect
- Rotating windmill animation
- Moving train and car

- Walking man animation
- Fish jumping in river
- Smooth animation using delay

CHAPTER 4

SCOPE OF THE PROJECT:

The scope of this project includes:

- Suitable for **students and beginners** learning C++
- Useful for understanding **console graphics**
- Can be used as an **academic mini-project**
- Can be extended with colors and sounds
- Helps in understanding real-time animation logic

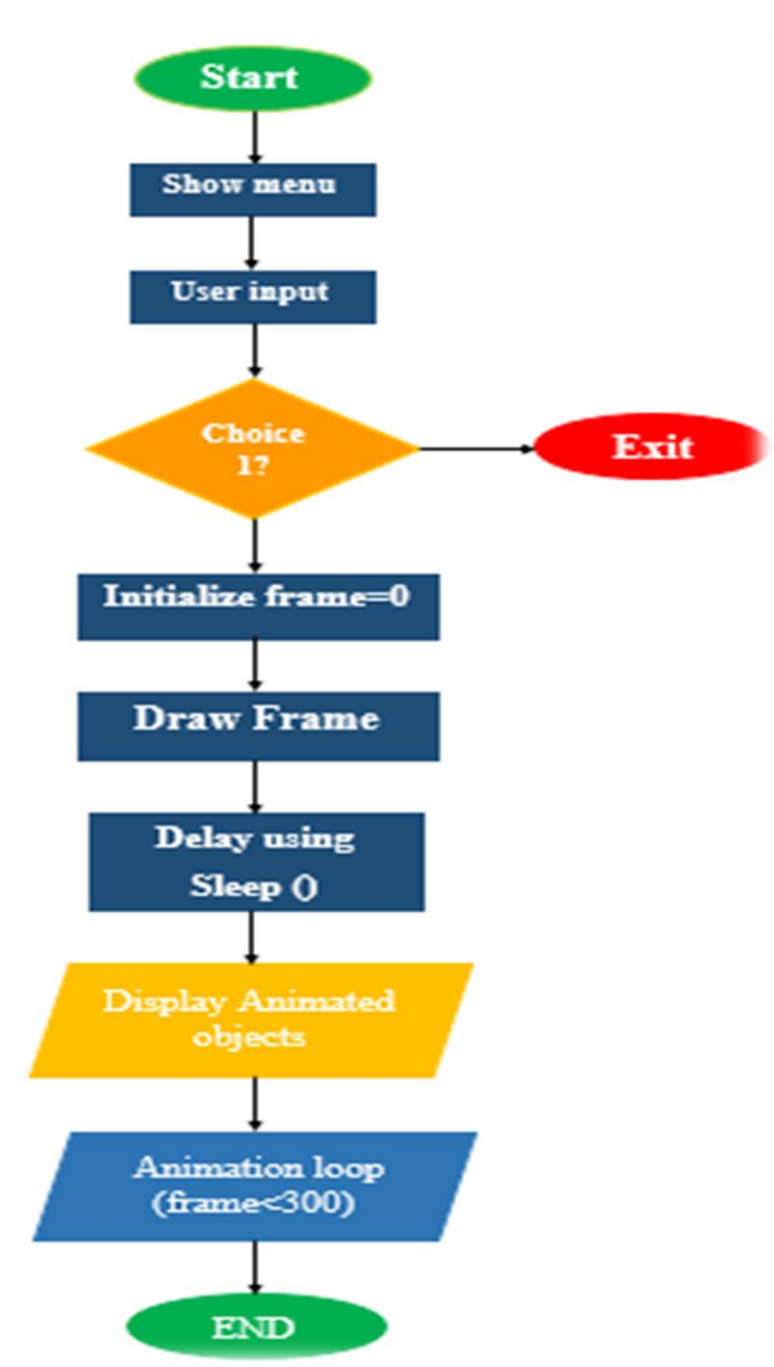
CHAPTER 5

ALGORITHM:

1. Start the program.
2. Display the menu.
3. Take user input.
4. If the user selects **Start Animation**:
    - Initialize frame counter.
    - Repeat until frame < 300:
        - Clear the console screen.
        - Calculate positions of all objects.
        - Draw the animated scene.
        - Apply delay using Sleep(100).
        - Increment frame.
    - Display completion message.
5. If the user selects **Exit**:
    - Display exit message.
6. End the program.

CHAPTER 6

FLOWCHART:



CHAPTER 7

TOOLS AND TECHNOLOGIES USED

Tools

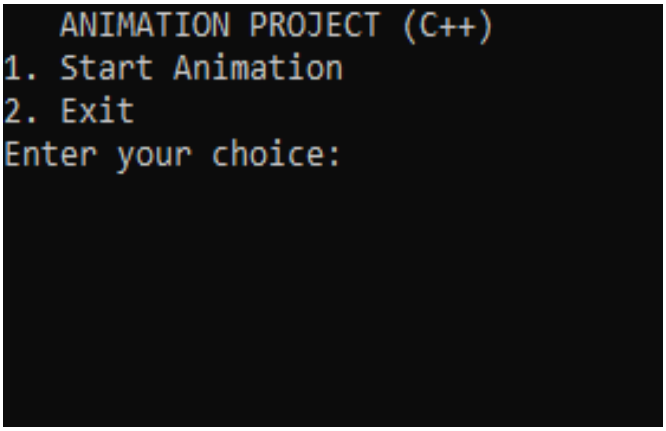
- Dev C++
- Visual Studio (optional)
- MS Word (for documentation)
- MS PowerPoint (for presentation)

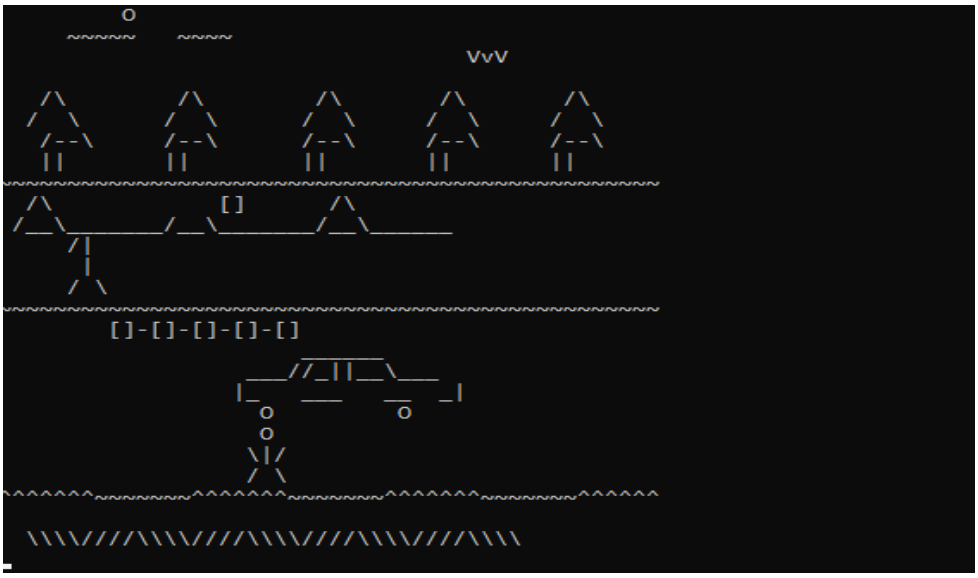
Technologies

- C++ Programming Language
- Console-based programming
- Windows API (windows.h)

CHAPTER 8

OUTPUT (SCREENSHOT):





CHAPTER 9

SOURCE CODE:

```
#include <iostream>
#include <windows.h>
#include <stdlib.h>
using namespace std;
void drawFrame(int frame)
{
        system("cls");
   // Speed control variables
   int sunPos   = (frame / 2) % 40;
   int cloudPos = (frame / 3) % 60;
   int birdPos  = (frame * 2) % 70;
   int carPos   = frame % 45;
```

```
int trainPos = (frame / 2) % 60;
int manPos   = frame % 40;
int windPos  = frame % 4;
int wavePos  = frame % 4;
int leafWave = frame % 2;
int lightOn  = frame % 4 < 2; // house lights blink
int fishJump = frame % 20;    // fish jumps sometimes
int grassWave = frame % 2;
// -------- SKY --------
for (int i = 0; i < sunPos; i++) cout << " ";
cout << " O " << endl; // Sun
for (int i = 0; i < cloudPos; i++) cout << " ";
cout << "~~~~~   ~~~~" << endl; // Clouds
for (int i = 0; i < birdPos; i++) cout << " ";
if (frame % 2 == 0)
   cout << "vVv" << endl;
else
   cout << "VvV" << endl;
cout << endl;
// -------- MOUNTAINS --------
cout << "   /\\       /\\       /\\       /\\       /\\      " << endl;
cout << "  /  \\     /  \\     /  \\     /  \\     /  \\     " << endl;
// -------- TREES --------
if (leafWave == 0)
   cout << "   /\\       /\\       /\\       /\\       /\\     " << endl;
```

```
else
    cout << "   /--\\     /--\\      /--\\     /--\\     /--\\   " << endl;
cout << "   ||       ||       ||       ||       ||      " << endl;
cout << "~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~" << endl;
// -------- HOUSE --------
if (lightOn)
    cout << "  /\\        " << "    []   " << "   /\\       " << endl;
else
    cout << "  /\\        " << "    --   " << "   /\\       " << endl;
cout << " /__\\_______/__\\_______/__\\______" << endl;

// -------- WINDMILL --------
if (windPos == 0)
    cout << "     |\\ " << endl;
else if (windPos == 1)
    cout << "     /| " << endl;
else if (windPos == 2)
    cout << "     -|- " << endl;
else
    cout << "     \\|/ " << endl;
cout << "      |  " << endl;
cout << "     / \\ " << endl;
cout << "~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~" << endl;
// -------- TRAIN IN BACKGROUND --------
for (int i = 0; i < trainPos; i++) cout << " ";
```

```
cout << "[]-[]-[]-[]-[]" << endl;


// -------- CAR --------
for (int i = 0; i < carPos; i++) cout << " ";
cout << "     _______" << endl;
for (int i = 0; i < carPos; i++) cout << " ";
cout << " ___//_||__\\___" << endl;
for (int i = 0; i < carPos; i++) cout << " ";
cout << "|_  ___   __  _|" << endl;
for (int i = 0; i < carPos; i++) cout << " ";
cout << "  O         O" << endl;
// -------- MAN WALKING --------
for (int i = 0; i < manPos; i++) cout << " ";
if (frame % 2 == 0)
        {
    cout << "  O  " << endl;
    for (int i = 0; i < manPos; i++) cout << " ";
    cout << " /|\\ " << endl;
    for (int i = 0; i < manPos; i++) cout << " ";
    cout << " / \\ " << endl;
} else
        {
    cout << "  O  " << endl;
    for (int i = 0; i < manPos; i++) cout << " ";
    cout << " \\|/ " << endl;
```

```
    for (int i = 0; i < manPos; i++) cout << " ";
    cout << " / \\ " << endl;
}
// -------- RIVER + FISH --------
if (wavePos == 0)
    cout << "~~~~~~~^^^^^^~~~~~~~^^^^^^~~~~~~~^^^^^^~~~~~~" << endl;
else if (wavePos == 1)
    cout << "^^^^^^~~~~~~~^^^^^^~~~~~~~^^^^^^~~~~~~~^^^^^^" << endl;
else if (wavePos == 2)
    cout << "~~~~~~~^^^^^^~~~~~~~^^^^^^~~~~~~~^^^^^^~~~~~~" << endl;
else
    cout << "^^^^^^~~~~~~~^^^^^^~~~~~~~^^^^^^~~~~~~~^^^^^^" << endl;

// Fish jump animation
if (fishJump > 10 && fishJump < 14)
        {
    for (int i = 0; i < fishJump; i++) cout << " ";
    cout << "><(((º>" << endl;
} else
        {
    cout << endl;
}
// -------- GRASS SWAY --------
if (grassWave == 0)
    cout << "  ///\\\\\\\\\///\\\\\\\\\///\\\\\\\\\///\\\\\\\\\///" << endl;
```

```
    else
        cout << "  \\\\\\\\/////\\\\\\\\/////\\\\\\\\/////\\\\\\\\/////\\\\\\\\" << endl;
}
int main()
{
    int choice;
    cout << "   ANIMATION PROJECT (C++)   \n";

    cout << "1. Start Animation\n";
    cout << "2. Exit\n";
    cout << "Enter your choice: ";
    cin >> choice;
    if (choice == 1)
    {
        for (int frame = 0; frame < 300; frame++)
        {
            drawFrame(frame);
            Sleep(100);
        }
        system("cls");
        cout << "\nAll 12 animations completed successfully!\n";
    }
    else
    {
        cout << "\nProgram exited without animation.\n";
```

```
    }
    return 0;
}
```

CHAPTER 10

Detailed Animation Explanation:

1**. Sun Movement:**
The sun moves horizontally across the top of the screen. The position is calculated by dividing the frame number and applying a modulo, ensuring a slow and smooth movement.

2. **Cloud Animation:**
Clouds move slower compared to the sun to give a realistic sky effect. They shift horizontally across the screen.

3**. Bird Animation:**
The birds flap their wings by alternating between `vVv` and `VvV` shapes. Their position changes faster to simulate flight.

4. **Mountains:**
Mountains are static ASCII structures placed below the sky. They do not move but help build a natural environment.

5. **Tree Animation:**
The leaves of the trees alternate between `/\` and `/--\` to create a waving effect.

6. **House Animation:**
The house has blinking windows. The window changes between `[]` (ON) and `--` (OFF) to simulate lighting.

7**. Windmill Animation:**
The windmill blades rotate using four different ASCII shapes: `|\`, `/|`, `-|-`, and `\|/`.

8**. Train Movement:**
The train moves in the background using ASCII compartments like `[]-[]-[]-[]`.

9. **Car Animation:**
The car moves smoothly across the screen with wheels represented by 'O'. The whole car shifts to the right each frame.

10. **Man Walking Animation:**
The man moves across the screen, and his arms alternate between `/|\` and `\|/` to show walking movement.

11. **River Waves:**
The river waves alternate between patterns of `~~~~~~~` and `^^^^^^^` to create flowing water motion.

12. **Fish Jump:**
The fish jumps between a small range of frames using ASCII such as `><(((º>`.

13**. Grass Swaying:**
The grass alternates between `////\\\\` and `\\\\////` to create a natural wind effect.

CHAPTER 11

Frame Calculation Logic:

The project makes heavy use of modulo (%) operations to animate movement. For example:
- (frame / 2) % 40 is used for sun movement.
- (frame / 3) % 60 is used for cloud movement.
- (frame * 2) % 70 is used for bird movement.
- frame % 45 is used for car movement.
- frame % 40 is used for man walking.

Modulo ensures the movement resets after reaching a certain limit, creating a looping animation.

CHAPTER 12

FUTURE ENHANCEMENTS:

- Add color using SetConsoleTextAttribute
- Add sound effects
- Improve animation smoothness
- Add more animated objects
- Convert console animation to graphics-based animation
- Make the program cross-platform

CHAPTER 13

REFERENCES:

- C++ Programming Language Documentation
- <iostream> Library
- <windows.h> Library
- Class Lecture Notes
- Online programming tutorials
- Programming textbooks

CHAPTER 14

CONCLUSION:

This project successfully demonstrates how animation can be created using **basic C++ programming concepts**. The program efficiently uses loops, conditions, and delays to simulate movement in a console environment.

The project enhanced understanding of animation logic and strengthened programming fundamentals. It is an excellent learning experience for beginners.

THE END